

Purpose

Evaluating whether offering an optional Summarize button helps desktop Chrome users comprehend long articles faster without sacrificing the quality. Two hypotheses are being tested.

Hypothesis 1: Reading Speed

Assumption: Normally 60% of readers pass the comprehension quiz within 3 mins.

H0: Offering the button does not increase the proportion of users who pass the quiz within 3 mins.

H1: offering the button increase comprehension by at least the minimum detectable effect of +3 percentage points to the users. (lifting passes from 60% to 63% or 30 extra successful readers per 1000 users)

Hypothesis 2: Comprehension Safety

Assumption: Normally 80% of readers pass when given the full 10-min window

H0: offering up the button reduces overall 10-min comprehension by 2 pp or more. (dropping total passes below 78%)

H1: Overall comprehension is preserved within our acceptable 2% safety margin.

Plan Of Action

- Participants read 1k-1.5k word article and completed 5 questions. Passing require at least 4 out of 5 correct.
- Randomization and Unit: Unique chrome profile ID with 1:1 allocation (50% split between control and treatment).
- Only each unique user's first attempt is analyzed to prevent skewness.
- Type I error was set to 5% and 90% detection power.
- Baselines: 60% timely comprehension (3mins) and 80% eventual comprehension (≤ 10 mins)

Distributions Used

- **Multinomial Distribution:** a standard coin flip has only 2 sides, but an article reader has 3 distinct ways their session can end.
 - o Fast Pass: pass the quiz in 3 mins or less
 - o Slow Pass: pass the quiz but takes between 3-10mins
 - o Unsuccessful: Fails the quiz, runs out of time or close the tab
- **Normal Distribution**
 - o Calculated whether 2% or 3% speed increase was a real victory or a random chance. (p value)
 - o Checking safety guardrail: established 95% worst-case boundary for accuracy. If the worst-case showed acc dropping by more than 2pp, the bell curve flagged it as unsafe.

Sample Size determination

- Dataset for speed test calculation: used a standard 2 sample comparison (z-test), detecting +3% speedup from the assumed 60% baseline with 90% power and 5% false alarm, it boiled down to 5527 users per group.
- Dataset for safety guardrail(comprehension): the assumed baseline was 80% and to make sure it does not drop by more than 2% required 6852 users. Proving 'no harm' against 2% limit required a larger sample than spotting 3% gain.
- Since both Gates must pass (speed and comprehension), the larger requirement governed the experiment and sample size was taken out to be 7000 users per group (treatment and control).
- At an intake of 1000 profiles per day, the test ran for 14 days.

Conclusion

- System approved it 88.34% of the time.
- Blocked dangerous features 100% of the time
- Ignored random noises: when button did nothing at all. All false alarm were kept down to 2.44% well within out type 1 error.